

DS Project Review-2

Government Grievance & Public Sentiment Analysis System

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Under the Guidance of

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Outline of Project Work

- Focus

- Introduction

- Literature Survey

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- Research Objectives

- Conclusion & Future Work

- References

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Problem Statement

Government grievance data contains valuable information about public issues, but raw complaint text is difficult to analyze directly.

- Citizen complaints may contain missing values, duplicate records and unnecessary information.
- Complaint text is unstructured, making it difficult to group complaints into major public-service categories.
- It is difficult to quickly determine whether citizen feedback is positive, negative or neutral.
- Complaint volumes and sentiment can change over time, so trend analysis is needed.
- A simple, interactive dashboard is needed to convert raw grievance data into understandable insights.

Objectives

- **Collect, clean and preprocess** government grievance data by handling missing values, duplicate records and unnecessary information.
- **Apply NLP and Machine Learning** to process citizen complaints and classify them into categories such as Roads, Water Supply, Electricity, Healthcare, Education and Transportation.
- **Perform Sentiment and Trend Analysis** to identify Positive, Negative and Neutral feedback and understand complaint patterns over time.
- **Develop an interactive Streamlit dashboard** to visualize complaint categories, sentiment distribution, trends and key insights.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Pang, Lee & Vaithyanathan	Thumbs up? Sentiment Classification using Machine Learning Techniques	Naive Bayes, Maximum Entropy, SVM	Machine learning can effectively classify text into sentiment categories.	Mainly focuses on movie reviews; government grievance-specific analysis can be explored.
Medhat, Hassan & Korashy	Sentiment Analysis Algorithms and Applications: A Survey	Lexicon-based and Machine Learning approaches	Provides an overview of different sentiment analysis techniques and applications.	Sentiment analysis can be affected by domain and language complexity.
Devlin et al	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	Transformer-based NLP	Contextual language models provide strong performance in text understanding.	Requires higher computational resources compared with basic ML approaches.

Problem Gap Identification

- Existing sentiment analysis systems mainly focus on identifying the **sentiment of text**.
- Existing grievance systems may focus on **collecting and managing complaints** without deeper sentiment analysis.
- Raw government grievance data requires **proper cleaning and preprocessing** before analysis.
- There is a need to combine **complaint classification and sentiment analysis**.
- Complaint frequency and sentiment should also be analyzed according to **time-based trends**.
- A simple interactive visualization can make these insights easier for users to understand.
- **Identified Gap**
- **CivicSense combines grievance analysis, complaint classification, sentiment analysis, trend analysis and interactive visualization into a single Data Science workflow.**

Innovation, Creativity, and Novelty

1. INNOVATION

Integrated Data Science: Combines data cleaning, NLP, Machine Learning and visualization in one system.

Automated Complaint Analysis: Automatically classifies citizen complaints into public-service categories and identifies their sentiment.

2. CREATIVITY

Citizen-Centric Analysis: Focuses specifically on real government grievances such as roads, water, electricity, healthcare, education and transportation.

Interactive Insights: Uses a Streamlit dashboard with charts, graphs, filters and statistics to make complex grievance data easy to understand.

3. NOVELTY

Unified Data Science Workflow: Combines **grievance analysis + complaint classification + sentiment analysis + trend analysis + visualization** in a single system.

Time-Based Insights: Goes beyond individual complaints by analyzing how complaint frequency and sentiment change over time.

Feasibility and Project Plan

Feasibility: Uses Python-based, widely available data-science libraries and publicly available grievance/sentiment datasets.

Phase	Week 1	Week 2	Week 3	Week 4	Week 5
Data Collection	✓	✓			
Cleaning & NLP		✓	✓		
Classification & Sentiment			✓	✓	
Trend Analysis & Dashboard				✓	✓
Testing, Insights & Finalization					✓

Overall Project Flow: Data Collection → Data Cleaning → Text Preprocessing → Complaint Classification → Sentiment Analysis → Trend Analysis → Dashboard → Insights